

Pharmacology

“Take me, I am the drug; take me, I am hallucinogenic.”

—Salvador Dali

“I was under medication when I made the decision not to burn the tapes.”

—Richard Nixon

“I wondher why ye can always read a doctor’s bill an’ ye niver can read his purscription.”

—Finley Peter Dunne

“Once you get locked into a serious drug collection, the tendency is to push it as far as you can.”

—Hunter S. Thompson

Preparation for questions on pharmacology is straightforward. Memorizing all the key drugs and their characteristics (eg, mechanisms, clinical use, and important side effects) is high yield. Focus on understanding the prototype drugs in each class. Avoid memorizing obscure derivatives. Learn the “classic” and distinguishing toxicities of the major drugs. Specific drug dosages or trade names are generally not testable. Reviewing associated biochemistry, physiology, and microbiology can be useful while studying pharmacology. There is a strong emphasis on ANS, CNS, antimicrobial, and cardiovascular agents as well as on NSAIDs. Much of the material is clinically relevant. We occasionally mention drugs that are no longer available in the US, but help illustrate high-yield pharmacologic or disease mechanisms. They are highlighted as being of historical significance and should not appear on the USMLE. However, recently approved drugs are fair game for the exam.

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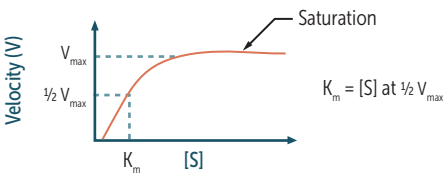
▶ PHARMACOLOGY—PHARMACOKINETICS & PHARMACODYNAMICS

Enzyme kinetics

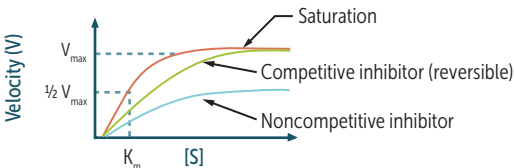
Michaelis-Menten kinetics

K_m is inversely related to the affinity of the enzyme for its substrate.
 V_{max} is directly proportional to the enzyme concentration.
Most enzymatic reactions follow a hyperbolic curve (ie, Michaelis-Menten kinetics); however, enzymatic reactions that exhibit a sigmoid curve usually indicate cooperative kinetics (eg, hemoglobin).

$[S]$ = concentration of substrate; V = velocity.

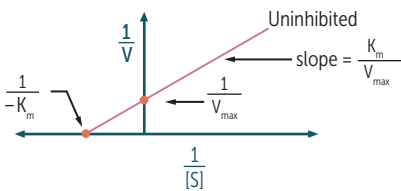


Effects of enzyme inhibition

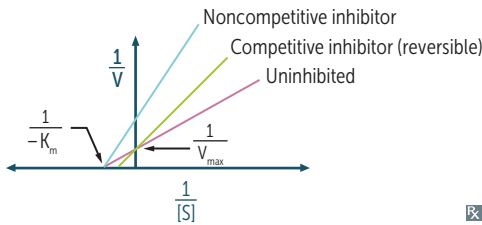


Lineweaver-Burk plot

↑ y-intercept, ↓ V_{max} .
The further to the right the x-intercept (ie, closer to zero), the greater the K_m and the lower the affinity.



Effects of enzyme inhibition



Reversible competitive inhibitors cross each other competitively, whereas noncompetitive inhibitors do not.

	Competitive inhibitors, reversible	Competitive inhibitors, irreversible	Noncompetitive inhibitors
Resemble substrate	Yes	Yes	No
Overcome by ↑ [S]	Yes	No	No
Bind active site	Yes	Yes	No
Effect on V_{max}	Unchanged	↓	↓
Effect on K_m	↑	Unchanged	Unchanged
Pharmacodynamics	↓ potency	↓ efficacy	↓ efficacy

Pharmacokinetics

Bioavailability (F)	Fraction of administered drug reaching systemic circulation unchanged. For an IV dose, F = 100%. Orally: F typically < 100% due to incomplete absorption and first-pass metabolism.				
Volume of distribution (V _d)	Theoretical volume occupied by the total amount of drug in the body relative to its plasma concentration. Apparent V _d of plasma protein-bound drugs can be altered by liver and kidney disease (↓ protein binding, ↑ V _d). Drugs may distribute in more than one compartment.				
	$V_d = \frac{\text{amount of drug in the body}}{\text{plasma drug concentration}}$				
	V _d	COMPARTMENT	DRUG TYPES		
	Low	Intravascular	Large/charged molecules; plasma protein bound		
	Medium	ECF	Small hydrophilic molecules		
	High	All tissues including fat	Small lipophilic molecules, especially if bound to tissue protein		
Clearance (CL)	The volume of plasma cleared of drug per unit time. Clearance may be impaired with defects in cardiac, hepatic, or renal function.				
	$CL = \frac{\text{rate of elimination of drug}}{\text{plasma drug concentration}} = V_d \times K_e \text{ (elimination constant)}$				
Half-life (t _{1/2})	The time required to change the amount of drug in the body by ½ during elimination. In first-order kinetics, a drug infused at a constant rate takes 4–5 half-lives to reach steady state. It takes 3.3 half-lives to reach 90% of the steady-state level.				
	$t_{1/2} = \frac{0.693 \times V_d}{CL} \text{ in first-order elimination}$		# of half-lives	1	2
			% remaining	50%	25%
				12.5%	6.25%

Dosage calculations

$$\text{Loading dose} = \frac{C_p \times V_d}{F}$$

$$\text{Maintenance dose} = \frac{C_p \times CL \times \tau}{F}$$

C_p = target plasma concentration at steady state
 τ = dosage interval (time between doses), if not administered continuously

In renal or liver disease, maintenance dose $\downarrow$ and loading dose is usually unchanged.
 Time to steady state depends primarily on $t_{1/2}$ and is independent of dose and dosing frequency.

Types of drug interactions

TERM	DEFINITION	EXAMPLE
Additive	Effect of substance A and B together is equal to the sum of their individual effects	Aspirin and acetaminophen
Permissive	Presence of substance A is required for the full effects of substance B	Cortisol on catecholamine responsiveness
Synergistic	Effect of substance A and B together is greater than the sum of their individual effects	Clopidogrel with aspirin
Tachyphylactic	Acute decrease in response to a drug after initial/repeated administration	Nitrates, niacin, phenylephrine, LSD, MDMA

Elimination of drugs

Zero-order elimination

Rate of elimination is constant regardless of C_p (ie, constant **amount** of drug eliminated per unit time). $C_p \downarrow$ linearly with time. Examples of drugs—**P**henytoin, **E**thanol, and **A**spirin (at high or toxic concentrations).

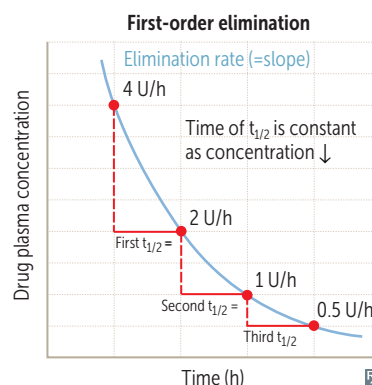
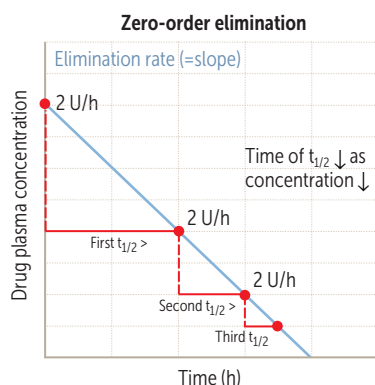
Capacity-limited elimination.

PEA. (A pea is round, shaped like the “0” in **zero-order**.)

First-order elimination

Rate of elimination is directly proportional to the drug concentration (ie, constant **fraction** of drug eliminated per unit time). $C_p \downarrow$ exponentially with time. Applies to most drugs.

Flow-dependent elimination.

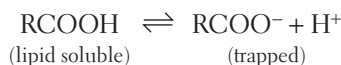


Urine pH and drug elimination

Ionized species are trapped in urine and cleared quickly. Neutral forms can be reabsorbed.

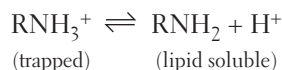
Weak acids

Examples: phenobarbital, methotrexate, aspirin (salicylates). Trapped in basic environments. Treat overdose with bicarbonate to alkalinize urine.



Weak bases

Example: amphetamines, TCAs. Trapped in acidic environments. Treat overdose with ammonium chloride to acidify urine.



Drug metabolism

Phase I

Reduction, oxidation, hydrolysis with cytochrome P-450 usually yield slightly polar, water-soluble metabolites (often still active).

Geriatric patients lose phase I first.

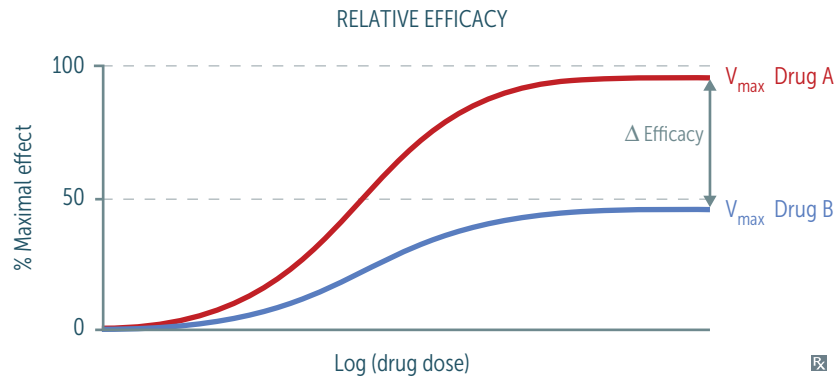
Phase II

Conjugation (**M**ethylation, **G**lucuronidation, **A**cetylation, **S**ulfation) usually yields very polar, inactive metabolites (renally excreted).

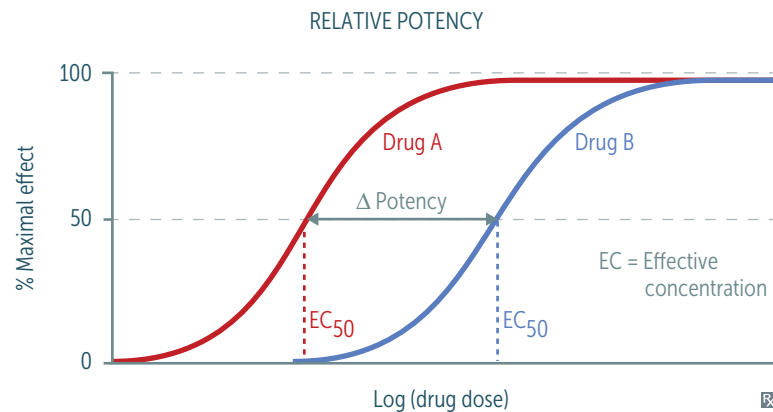
Geriatric patients have **More GAS** (phase II). Patients who are slow acetylators have $\uparrow$ side effects from certain drugs because of $\downarrow$ rate of metabolism.

Efficacy vs potency**Efficacy**

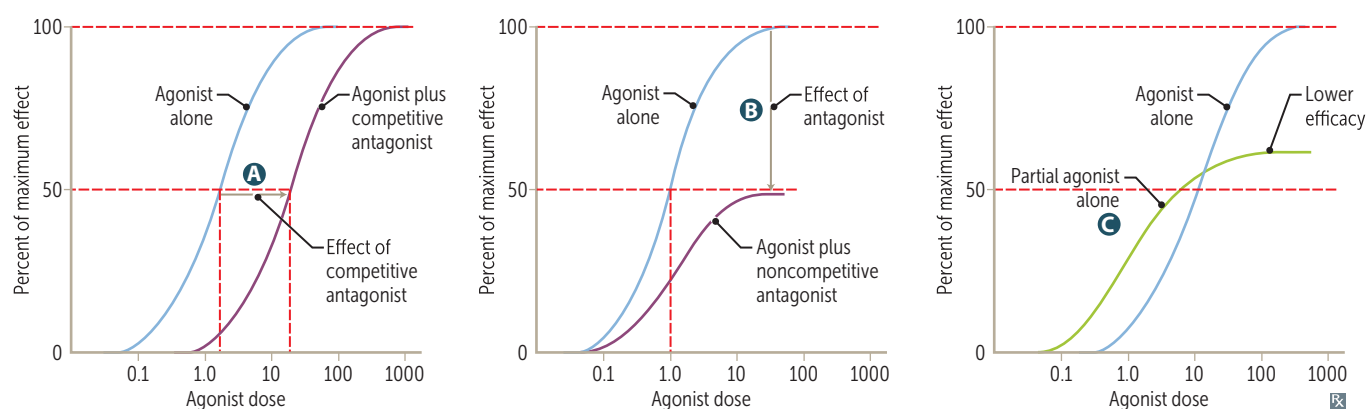
Maximal effect a drug can produce. Represented by the y-value ($V_{\max}$). $\uparrow$ y-value = $\uparrow V_{\max}$ = $\uparrow$ efficacy. Unrelated to potency (ie, efficacious drugs can have high or low potency). Partial agonists have less efficacy than full agonists.

**Potency**

Amount of drug needed for a given effect. Represented by the x-value (EC_{50}). Left shifting = $\downarrow EC_{50}$ = $\uparrow$ potency = $\downarrow$ drug needed. Unrelated to efficacy (ie, potent drugs can have high or low efficacy).



Receptor binding



AGONIST WITH	EFFECT	EXAMPLE
A Competitive antagonist	Shifts curve right (↓ potency), no change in efficacy. Can be overcome by ↑ the concentration of agonist substrate.	Diazepam (agonist) + flumazenil (competitive antagonist) on GABA receptor.
B Noncompetitive antagonist	Shifts curve down (↓ efficacy). Cannot be overcome by ↑ agonist substrate concentration.	Norepinephrine (agonist) + phenoxybenzamine (noncompetitive antagonist) on α-receptors.
C Partial agonist (alone)	Acts at same site as full agonist, but with lower maximal effect (↓ efficacy). Potency is an independent variable.	Morphine (full agonist) vs buprenorphine (partial agonist) at opioid μ-receptors.

Therapeutic index

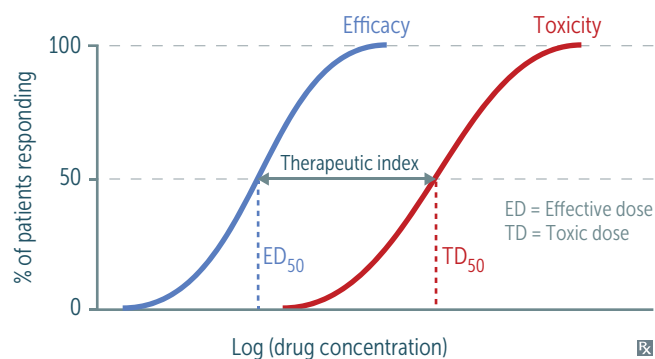
Measurement of drug safety.

$$\frac{TD_{50}}{ED_{50}} = \frac{\text{median toxic dose}}{\text{median effective dose}}$$

Therapeutic window—dosage range that can safely and effectively treat disease.

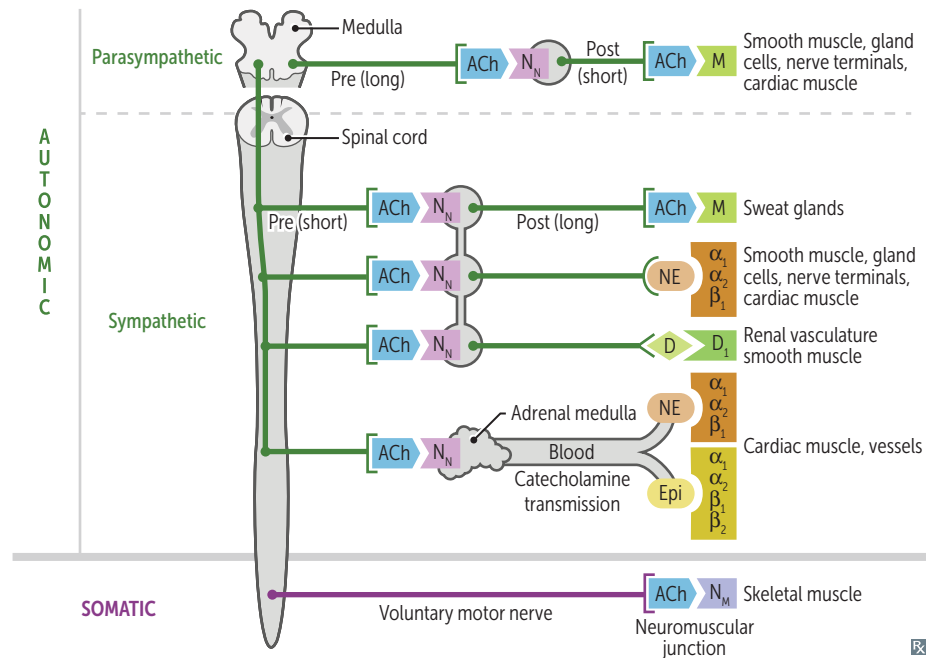
TITE: Therapeutic Index = TD_{50} / ED_{50} .

Safer drugs have higher TI values. Drugs with lower TI values frequently require monitoring (eg, **Warfarin**, **Theophylline**, **Digoxin**, **Lithium**; **Warning! These Drugs are Lethal!**). LD₅₀ (lethal median dose) often replaces TD₅₀ in animal studies.



► PHARMACOLOGY—AUTONOMIC DRUGS

Central and peripheral nervous system



Adrenal medulla is directly innervated by preganglionic sympathetic fibers.

Sweat glands are part of the sympathetic pathway but are innervated by cholinergic fibers.

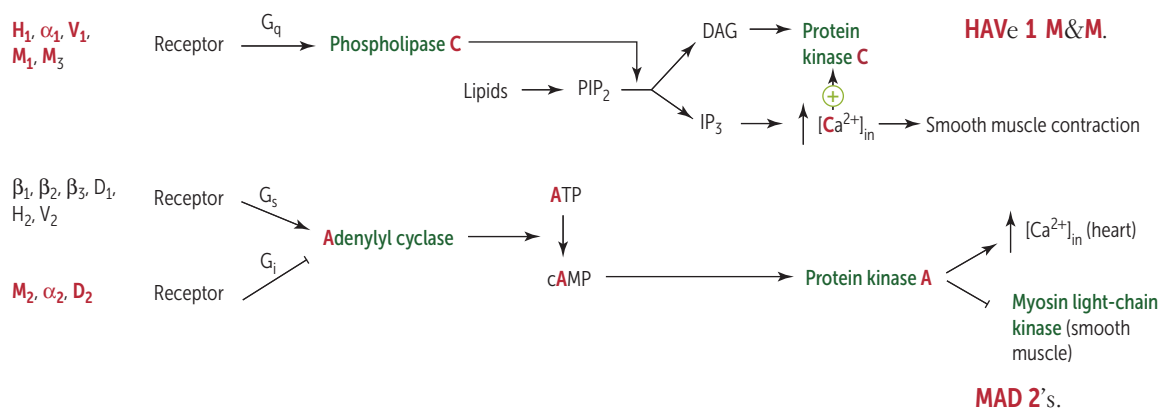
Acetylcholine receptors

Nicotinic ACh receptors are ligand-gated Na^+/K^+ channels. Two subtypes: N_N (found in autonomic ganglia, adrenal medulla) and N_M (found in neuromuscular junction of skeletal muscle). Muscarinic ACh receptors are G-protein-coupled receptors that usually act through 2nd messengers. 5 subtypes: M_{1-5} found in heart, smooth muscle, brain, exocrine glands, and on sweat glands (cholinergic sympathetic).

G-protein-linked second messengers

RECEPTOR	G-PROTEIN CLASS	MAJOR FUNCTIONS
Sympathetic		
α_1	q	↑ vascular smooth muscle contraction, ↑ pupillary dilator muscle contraction (mydriasis), ↑ intestinal and bladder sphincter muscle contraction
α_2	i	↓ sympathetic (adrenergic) outflow, ↓ insulin release, ↓ lipolysis, ↑ platelet aggregation, ↓ aqueous humor production
β_1	s	↑ heart rate, ↑ contractility (one heart), ↑ renin release, ↑ lipolysis
β_2	s	Vasodilation, bronchodilation (two lungs), ↑ lipolysis, ↑ insulin release, ↓ uterine tone (tocolysis), ciliary muscle relaxation, ↑ aqueous humor production
β_3	s	↑ lipolysis, ↑ thermogenesis in skeletal muscle, ↑ bladder relaxation
Parasympathetic		
M_1	q	Mediates higher cognitive functions, stimulates enteric nervous system
M_2	i	↓ heart rate and contractility of atria
M_3	q	↑ exocrine gland secretions (eg, lacrimal, sweat, salivary, gastric acid), ↑ gut peristalsis, ↑ bladder contraction, bronchoconstriction, ↑ pupillary sphincter muscle contraction (miosis), ciliary muscle contraction (accommodation), ↑ insulin release
Dopamine		
D_1	s	Relaxes renal vascular smooth muscle, activates direct pathway of striatum
D_2	i	Modulates transmitter release, especially in brain, inhibits indirect pathway of striatum
Histamine		
H_1	q	↑ nasal and bronchial mucus production, ↑ vascular permeability, contraction of bronchioles, pruritus, pain
H_2	s	↑ gastric acid secretion
Vasopressin		
V_1	q	↑ vascular smooth muscle contraction
V_2	s	↑ H_2O permeability and reabsorption in collecting tu bules of kidney

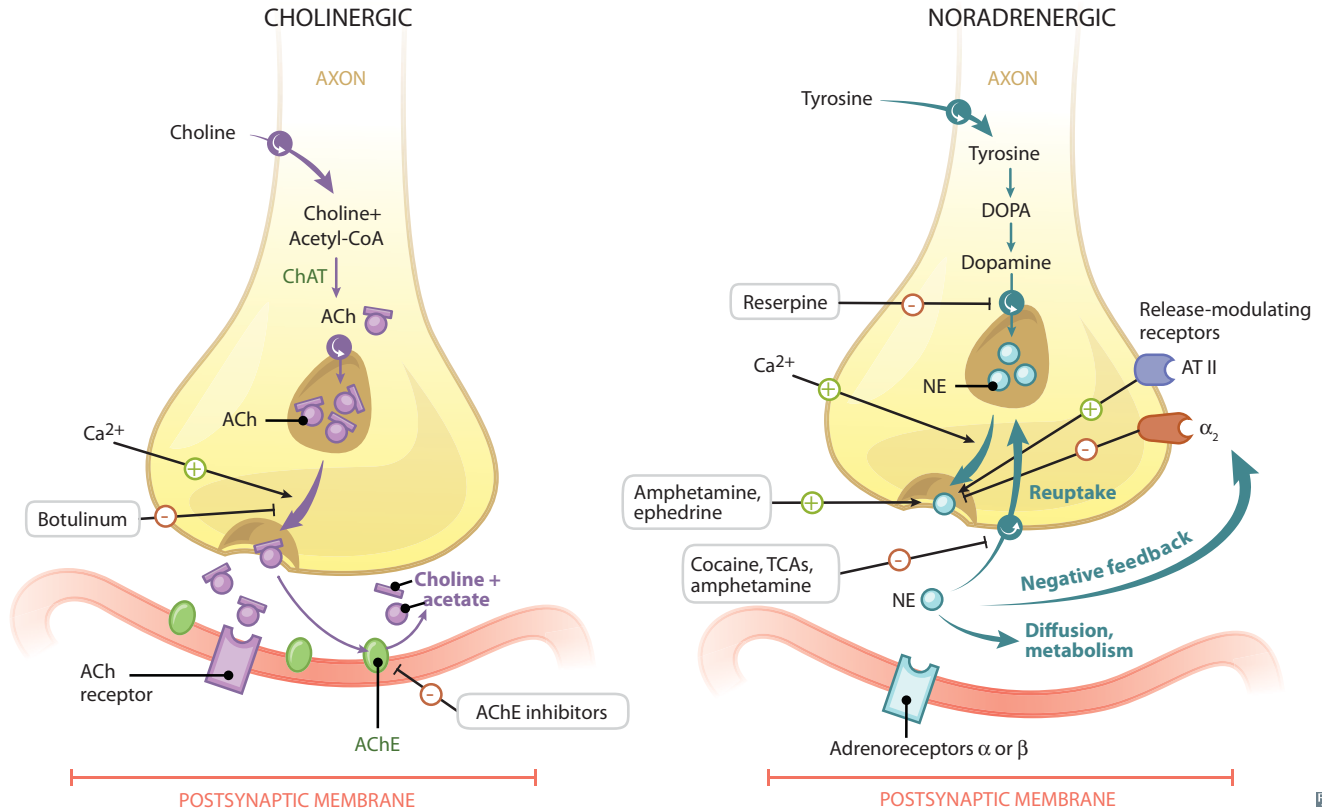
“After **q**isses (kisses), you get a **q**iq (kick) out of **s**iq (sick) **sqs** (super kinky sex).”



Autonomic drugs

Release of norepinephrine from a sympathetic nerve ending is modulated by NE itself, acting on presynaptic α_2 -autoreceptors → negative feedback.

Amphetamines use the NE transporter (NET) to enter the presynaptic terminal, where they utilize the vesicular monoamine transporter (VMAT) to enter neurosecretory vesicles. This displaces NE from the vesicles. Once NE reaches a concentration threshold within the presynaptic terminal, the action of NET is reversed, and NE is expelled into the synaptic cleft, contributing to the characteristics and effects of ↑ NE observed in patients taking amphetamines.



Cholinomimetic agents

DRUG	ACTION	APPLICATIONS
Direct agonists		
Bethanechol	Activates b owel and b ladder smooth muscle; resistant to AChE, no nicotinic activity. “ Bethany , call (bethanechol) me to activate your b owels and b ladder.”	Postoperative ileus, neurogenic ileus, urinary retention
Carbachol	C arbon copy of a cetyl ch oline (but resistant to AChE).	Constricts pupil and relieves intraocular pressure in open-angle glaucoma
Methacholine	Stimulates m uscarinic receptors in airway when inhaled.	Challenge test for diagnosis of asthma
Pilocarpine	Contracts ciliary muscle of eye (open-angle glaucoma), pupillary sphincter (closed-angle glaucoma); resistant to AChE, can cross blood-brain barrier (tertiary amine). “You cry, drool, and sweat on your p illow.”	Potent stimulator of sweat, tears, and saliva Open-angle and closed-angle glaucoma, xerostomia (Sjögren syndrome)
Indirect agonists (anticholinesterases)		
Galantamine, donepezil, rivastigmine	↑ ACh.	Alzheimer disease (Alzheimer patients g allantly swim d own the r iver)
Edrophonium	↑ ACh.	Historically used to diagnose myasthenia gravis; replaced by anti-AChR Ab (anti-acetylcholine receptor antibody) test.
Neostigmine	↑ ACh. Neo CNS = No CNS penetration (quaternary amine).	Postoperative and neurogenic ileus and urinary retention, myasthenia gravis, reversal of neuromuscular junction blockade (postoperative).
Physostigmine	↑ ACh. Phys ostigmine “ phy xes” atropine overdose.	Antidote for anticholinergic toxicity; ph reely (freely) crosses blood-brain barrier → CNS (tertiary amine).
Pyridostigmine	↑ ACh; ↑ muscle strength. Pyrid ostigmine gets rid of my asthenia g avis.	Myasthenia gravis (long acting); does not penetrate CNS (quaternary amine).

Note: With all cholinomimetic agents, watch for exacerbation of COPD, asthma, and peptic ulcers when giving to susceptible patients.

Cholinesterase inhibitor poisoning

Often due to organophosphates, such as parathion, that **irreversibly** inhibit AChE. Causes **D**iarrhea, **U**rination, **M**iosis, **B**ronchospasm, **B**radycardia, **E**xcitation of skeletal muscle and CNS, **L**acrimation, **S**weating, and **S**alivation. May lead to respiratory failure if untreated.

DUMBBELSS.

Organophosphates are often components of insecticides; poisoning usually seen in farmers. Antidote—atropine (competitive inhibitor) + pralidoxime (regenerates AChE if given early).

Muscarinic antagonists

DRUGS	ORGAN SYSTEMS	APPLICATIONS
Atropine, homatropine, tropicamide	Eye	Produce mydriasis and cycloplegia.
Benztropine, trihexyphenidyl	CNS	Parkinson disease (“ park my Benz ”). Acute dystonia.
Glycopyrrolate	GI, respiratory	Parenteral: preoperative use to reduce airway secretions. Oral: drooling, peptic ulcer.
Hyoscyamine, dicyclomine	GI	Antispasmodics for irritable bowel syndrome.
Ipratropium, tiotropium	Respiratory	COPD, asthma (“ I pray I can breathe soon!”).
Oxybutynin, solifenacin, tolterodine	Genitourinary	Reduce bladder spasms and urge urinary incontinence (overactive bladder).
Scopolamine	CNS	Motion sickness.

Atropine Muscarinic antagonist. Used to treat bradycardia and for ophthalmic applications.

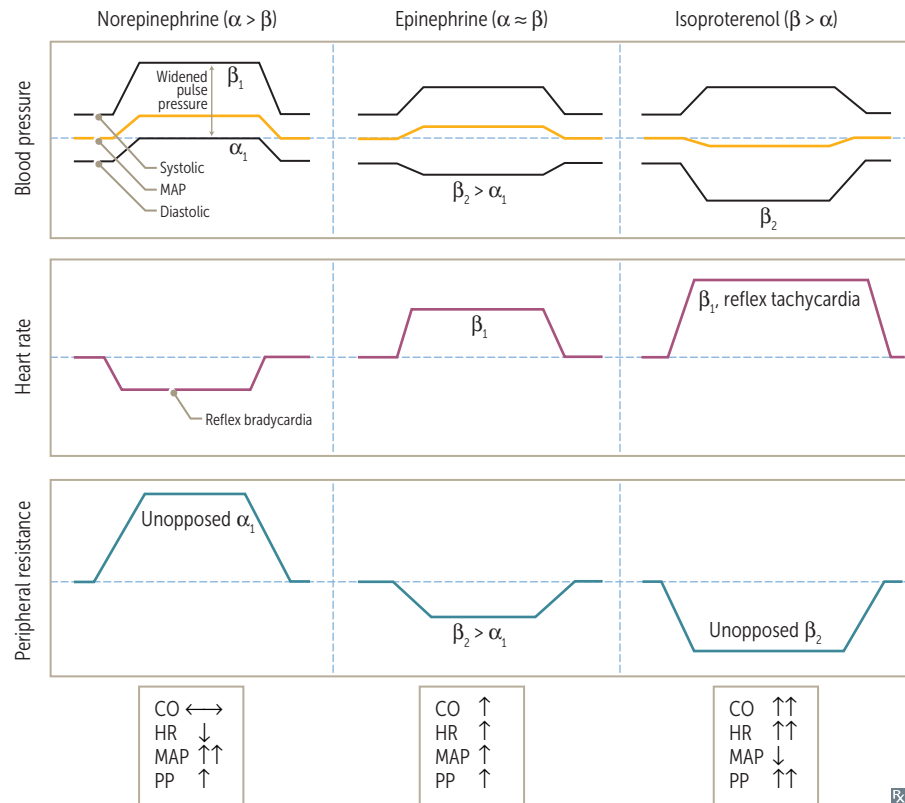
ORGAN SYSTEM	ACTION	NOTES
Eye	↑ pupil dilation, cycloplegia	Blocks DUMBB eLSS in cholinesterase inhibitor poisoning. Does not block excitation of skeletal muscle and CNS (mediated by nicotinic receptors).
Airway	↓ secretions	
Stomach	↓ acid secretion	
Gut	↓ motility	
Bladder	↓ urgency in cystitis	
ADVERSE EFFECTS	↑ body temperature (due to ↓ sweating); rapid pulse; dry mouth; dry, flushed skin; cycloplegia; constipation; disorientation Can cause acute angle-closure glaucoma in elderly (due to mydriasis), urinary retention in men with prostatic hyperplasia, and hyperthermia in infants.	Side effects: Hot as a hare Dry as a bone Red as a beet Blind as a bat Mad as a hatter Jimson weed (<i>Datura</i>) → gardener’s pupil (mydriasis due to plant alkaloids)

Sympathomimetics

DRUG	ACTION	APPLICATIONS
Direct sympathomimetics		
Albuterol, salmeterol	$\beta_2 > \beta_1$	Albuterol for acute asthma or COPD. Salmeterol for long-term asthma or COPD control.
Dobutamine	$\beta_1 > \beta_2, \alpha$	Heart failure (HF) (inotropic > chronotropic), cardiac stress testing.
Dopamine	$D_1 = D_2 > \beta > \alpha$	Unstable bradycardia, HF, shock; inotropic and chronotropic effects at lower doses due to β effects; vasoconstriction at high doses due to α effects.
Epinephrine	$\beta > \alpha$	Anaphylaxis, asthma, open-angle glaucoma; α effects predominate at high doses. Significantly stronger effect at β_2 -receptor than norepinephrine.
Fenoldopam	D_1	Postoperative hypertension, hypertensive crisis. Vasodilator (coronary, peripheral, renal, and splanchnic). Promotes natriuresis. Can cause hypotension and tachycardia.
Isoproterenol	$\beta_1 = \beta_2$	Electrophysiologic evaluation of tachyarrhythmias. Can worsen ischemia.
Midodrine	α_1	Autonomic insufficiency and postural hypotension. May exacerbate supine hypertension.
Mirabegron	β_3	Urinary urge incontinence or overactive bladder.
Norepinephrine	$\alpha_1 > \alpha_2 > \beta_1$	Hypotension, septic shock.
Phenylephrine	$\alpha_1 > \alpha_2$	Hypotension (vasoconstrictor), ocular procedures (mydriatic), rhinitis (decongestant).
Indirect sympathomimetics		
Amphetamine	Indirect general agonist, reuptake inhibitor, also releases stored catecholamines	Narcolepsy, obesity, ADHD.
Cocaine	Indirect general agonist, reuptake inhibitor	Causes vasoconstriction and local anesthesia. Never give β -blockers if cocaine intoxication is suspected (can lead to unopposed α_1 activation and extreme hypertension).
Ephedrine	Indirect general agonist, releases stored catecholamines	Nasal decongestion (pseudoephedrine), urinary incontinence, hypotension.

Norepinephrine vs isoproterenol

NE $\uparrow$ systolic and diastolic pressures as a result of α_1 -mediated vasoconstriction $\rightarrow \uparrow$ mean arterial pressure $\rightarrow$ reflex bradycardia. However, isoproterenol (rarely used) has little α effect but causes β_2 -mediated vasodilation, resulting in $\downarrow$ mean arterial pressure and $\uparrow$ heart rate through β_1 and reflex activity.

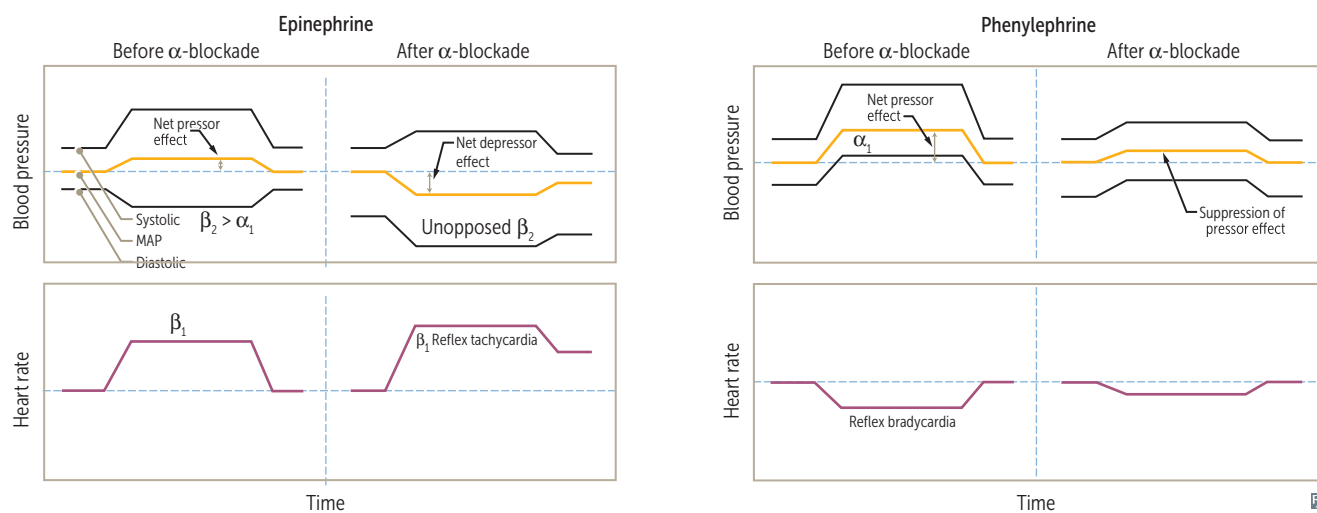


Sympatholytics (α_2 -agonists)

DRUG	APPLICATIONS	ADVERSE EFFECTS
Clonidine, guanfacine	Hypertensive urgency (limited situations), ADHD, Tourette syndrome	CNS depression, bradycardia, hypotension, respiratory depression, miosis, rebound hypertension with abrupt cessation
α-methyldopa	Hypertension in pregnancy	Direct Coombs $\oplus$ hemolysis, SLE-like syndrome

α -blockers

DRUG	APPLICATIONS	ADVERSE EFFECTS
Nonselective		
Phenoxybenzamine	Irreversible. Pheochromocytoma (used preoperatively) to prevent catecholamine (hypertensive) crisis	Orthostatic hypotension, reflex tachycardia
Phentolamine	Reversible. Give to patients on MAO inhibitors who eat tyramine-containing foods	
α_1 selective (-osin ending)		
Prazosin, terazosin, doxazosin, tamsulosin	Urinary symptoms of BPH; PTSD (prazosin); hypertension (except tamsulosin)	1st-dose orthostatic hypotension, dizziness, headache
α_2 selective		
Mirtazapine	Depression	Sedation, $\uparrow$ serum cholesterol, $\uparrow$ appetite

Effects of α -blocker (eg, phentolamine) on BP responses to epinephrine and phenylephrine

Epinephrine response exhibits reversal of mean arterial pressure from a net increase (the α response) to a net decrease (the β_2 response).

Phenylephrine response is suppressed but not reversed because it is a “pure” α -agonist (lacks β -agonist properties).

β-blockers

Acebutolol, atenolol, betaxolol, bisoprolol, carvedilol, esmolol, labetalol, metoprolol, nadolol, nebivolol, pindolol, propranolol, timolol.

APPLICATION	ACTIONS	NOTES/EXAMPLES
Angina pectoris	↓ heart rate and contractility, resulting in ↓ O ₂ consumption	
Myocardial infarction	↓ mortality	
Supraventricular tachycardia	↓ AV conduction velocity (class II antiarrhythmic)	Metoprolol, esmolol
Hypertension	↓ cardiac output, ↓ renin secretion (due to β ₁ -receptor blockade on JGA cells)	
Heart failure	↓ mortality (bisoprolol, carvedilol, metoprolol)	
Glaucoma	↓ production of aqueous humor	Timolol
Variceal bleeding	↓ hepatic venous pressure gradient and portal hypertension	Nadolol, propranolol
ADVERSE EFFECTS	Erectile dysfunction, cardiovascular adverse effects (bradycardia, AV block, HF), CNS adverse effects (seizures, sedation, sleep alterations), dyslipidemia (metoprolol), and asthma/COPD exacerbations	Use with caution in cocaine users due to risk of unopposed α-adrenergic receptor agonist activity
SELECTIVITY	β ₁ -selective antagonists (β ₁ > β ₂)— a cebutolol (partial agonist), a tenolol, b etaxolol, b isoprolol, e smolol, m etoprolol	Selective antagonists mostly go from A to M (β ₁ with 1 st half of alphabet)
	Nonselective antagonists (β ₁ = β ₂)— n adolol, p indolol (partial agonist), p ropranolol, t imolol	Nonselective antagonists mostly go from N to Z (β ₂ with 2 nd half of alphabet)
	Nonselective α- and β-antagonists—carvedilol, labetalol	Nonselective α- and β-antagonists have modified suffixes (instead of “-olol”)
	Nebivolol combines cardiac-selective β ₁ -adrenergic blockade with stimulation of β ₃ -receptors (activate nitric oxide synthase in the vasculature and ↓ SVR)	

Ingested seafood toxins

TOXIN	SOURCE	ACTION	SYMPTOMS	TREATMENT
Tetrodotoxin	Pufferfish.	Highly potent toxin; binds fast voltage-gated Na ⁺ channels in cardiac/nerve tissue, preventing depolarization.	Nausea, diarrhea, paresthesias, weakness, dizziness, loss of reflexes.	Supportive.
Ciguatoxin	Reef fish such as barracuda, snapper, and moray eel.	Opens Na ⁺ channels, causing depolarization.	Nausea, vomiting, diarrhea; perioral numbness; reversal of hot and cold sensations; bradycardia, heart block, hypotension.	Supportive.
Histamine (scombroid poisoning)	Spoiled dark-meat fish such as tuna, mahi-mahi, mackerel, and bonito.	Bacterial histidine decarboxylase converts histidine to histamine. Frequently misdiagnosed as fish allergy.	Mimics anaphylaxis: acute burning sensation of mouth, flushing of face, erythema, urticaria, itching. May progress to bronchospasm, angioedema, hypotension.	Antihistamines. Albuterol and epinephrine if needed.

Beers criteria

Widely used criteria developed to reduce potentially inappropriate prescribing and harmful polypharmacy in the geriatric population. Includes > 50 medications that should be avoided in elderly patients due to ↓ efficacy and/or ↑ risk of adverse events. Examples include:

- Anticholinergics, antihistamines, antidepressants, benzodiazepines, opioids (↑ risk of delirium, sedation, falls, constipation, urinary retention)
- α-blockers (↑ risk of hypotension)
- PPIs (↑ risk of *C difficile* infection)
- NSAIDs (↑ risk of GI bleeding, especially with concomitant anticoagulation)

► PHARMACOLOGY—TOXICITIES AND SIDE EFFECTS

Specific toxicity treatments

TOXIN	TREATMENT
Acetaminophen	N-acetylcysteine (replenishes glutathione)
AChE inhibitors, organophosphates	Atropine > pralidoxime
Antimuscarinic, anticholinergic agents	Physostigmine, control hyperthermia
Arsenic	Dimercaprol, succimer
Benzodiazepines	Flumazenil
β-blockers	Atropine, glucagon
Carbon monoxide	100% O ₂ , hyperbaric O ₂
Copper	Penicillamine , trientine (Copper penny)
Cyanide	Nitrite + thiosulfate, hydroxocobalamin
Digitalis (digoxin)	Anti-dig Fab fragments
Heparin	Protamine sulfate
Iron	De fer oxamine, de fer asirox, de fer iprone
Lead	EDTA, dimercaprol, succimer, penicillamine
Mercury	Di mer caprol, suc cimer
Methanol, ethylene glycol (antifreeze)	Fomepizole > ethanol, dialysis
Methemoglobin	Meth ylene blue, vitamin C
OpiOids	Nal OxO ne
Salicylates	NaHCO ₃ (alkalinize urine), dialysis
TCAs	NaHCO ₃
Warfarin	Vitamin K (delayed effect), fresh frozen plasma (immediate)

Drug reactions—cardiovascular

DRUG REACTION	CAUSAL AGENTS
Coronary vasospasm	Amphetamines, cocaine, ergot alkaloids, sumatriptan
Cutaneous flushing	V ancomycin, A denosine, N iacin, N itrates, Ca ²⁺ channel blockers, E chinocandins (VANNCE)
Dilated cardiomyopathy	Anthracyclines (eg, doxorubicin, daunorubicin); prevent with dexrazoxane
Torsades de pointes	Anti A rrhythmics (class IA, III), anti B iotics (eg, macrolides), anti“ C ”ychotics (eg, haloperidol), anti D epressants (eg, TCAs), anti E metics (eg, ondansetron) (ABCDE)

Drug reactions—endocrine/reproductive

DRUG REACTION	CAUSAL AGENTS	NOTES
Adrenocortical insufficiency	HPA suppression 2° to glucocorticoid withdrawal	
Diabetes insipidus	Lithium, demeclocycline	
Hot flashes	Tamoxifen, clomiphene	
Hyperglycemia	Tacrolimus, Protease inhibitors, Niacin, HCTZ, Corticosteroids	Taking Pills Necessitates Having blood Checked
Hypothyroidism	Lithium, amiodarone, sulfonamides	
SIADH	Carbamazepine, Cyclophosphamide, SSRIs	Can't Concentrate Serum Sodium

Drug reactions—gastrointestinal

DRUG REACTION	CAUSAL AGENTS	NOTES
Acute cholestatic hepatitis, jaundice	Erythromycin	
Diarrhea	Acamprosate, acarbose, cholinesterase inhibitors, colchicine, erythromycin, ezetimibe, metformin, misoprostol, orlistat, pramlintide, quinidine, SSRIs	
Focal to massive hepatic necrosis	Halothane, <i>Amanita phalloides</i> (death cap mushroom), Valproic acid, Acetaminophen	Liver “HAVA _c ”
Hepatitis	Rifampin, isoniazid, pyrazinamide, statins, fibrates	
Pancreatitis	Didanosine, Corticosteroids, Alcohol, Valproic acid, Azathioprine, Diuretics (furosemide, HCTZ)	Drugs Causing A Violent Abdominal Distress
Pill-induced esophagitis	Bisphosphonates, ferrous sulfate, NSAIDs, potassium chloride, tetracyclines	Caustic effect minimized with upright posture and adequate water ingestion.
Pseudomembranous colitis	Ampicillin, cephalosporins, clindamycin, fluoroquinolones	Antibiotics predispose to superinfection by resistant <i>C difficile</i>

Drug reactions—hematologic

DRUG REACTION	CAUSAL AGENTS	NOTES
Agranulocytosis	Clozapine, Carbamazepine, Propylthiouracil, Methimazole, Colchicine, Ganciclovir	Can Cause Pretty Major Collapse of Granulocytes
Aplastic anemia	Carbamazepine, Methimazole, NSAIDs, Benzene, Chloramphenicol, Propylthiouracil	Can't Make New Blood Cells Properly
Direct Coombs-positive hemolytic anemia	Methyldopa, penicillin	
Drug reaction with eosinophilia and systemic symptoms (DRESS)	Allopurinol, anticonvulsants, antibiotics, sulfa drugs	Potentially fatal delayed hypersensitivity reaction. Latency period (2–8 weeks) followed by fever, morbilliform skin rash, and frequent multiorgan involvement. Treatment: withdrawal of offending drug, corticosteroids.
Gray baby syndrome	Chloramphenicol	
Hemolysis in G6PD deficiency	Isoniazid, Sulfonamides, Dapsone, Primaquine, Aspirin, Ibuprofen, Nitrofurantoin	Hemolysis IS D PAIN
Megaloblastic anemia	Hydroxyurea, Phenytoin, Methotrexate, Sulfa drugs	You're having a mega blast with PMS
Thrombocytopenia	Heparin	
Thrombotic complications	Combined oral contraceptives, hormone replacement therapy, SERMs (eg, tamoxifen, raloxifene, clomiphene)	Estrogen-mediated side effect

Drug reactions—musculoskeletal/skin/connective tissue

DRUG REACTION	CAUSAL AGENTS	NOTES
Fat redistribution	Protease inhibitors, Glucocorticoids	Fat PiG
Gingival hyperplasia	Phenytoin, Ca ²⁺ channel blockers, cyclosporine	
Hyperuricemia (gout)	Pyrazinamide, Thiazides, Furosemide, Niacin, Cyclosporine	Painful Tophi and Feet Need Care
Myopathy	Statins, fibrates, niacin, colchicine, daptomycin, hydroxychloroquine, interferon- α , penicillamine, glucocorticoids	
Osteoporosis	Corticosteroids, depot medroxyprogesterone acetate, GnRH agonists, aromatase inhibitors, anticonvulsants, heparin	
Photosensitivity	Sulfonamides, Amiodarone, Tetracyclines, 5-FU	SAT For Photo
Rash (Stevens-Johnson syndrome)	Anti-epileptic drugs (especially lamotrigine), allopurinol, sulfa drugs, penicillin	Steven Johnson has epileptic allergy to sulfa drugs and penicillin
SLE-like syndrome	Sulfa drugs, Hydralazine, Isoniazid, Procainamide, Phenytoin, Etanercept	Having lupus is "SHIPP-E"
Teeth discoloration	Tetracyclines	Teethracyclines
Tendonitis, tendon rupture, and cartilage damage	Fluoroquinolones	

Drug reactions—neurologic

DRUG REACTION	CAUSAL AGENTS	NOTES
Cinchonism	Quinidine, quinine	Can present with tinnitus, hearing/vision loss, psychosis, and cognitive impairment
Parkinson-like syndrome	Antipsychotics, R eserpine, M etoclopramide	Cogwheel rigidity of ARM
Seizures	Isoniazid (vitamin B ₆ deficiency), B upropion, I mipenem/cilastatin, T ramadol, E nflurane	With seizures , I BITE my tongue
Tardive dyskinesia	Antipsychotics, metoclopramide	

Drug reactions—renal/genitourinary

DRUG REACTION	CAUSAL AGENTS	NOTES
Fanconi syndrome	Cisplatin, ifosfamide, expired tetracyclines, tenofovir	
Hemorrhagic cystitis	Cyclophosphamide, ifosfamide	Prevent by coadministering with mesna
Interstitial nephritis	Penicillins, furosemide, NSAIDs, proton pump inhibitors, sulfa drugs	

Drug reactions—respiratory

DRUG REACTION	CAUSAL AGENTS	NOTES
Dry cough	ACE inhibitors	
Pulmonary fibrosis	M ethotrexate, N itrofurantoin, C armustine, B leomycin, B usulfan, A miodarone	My Nose Cannot Breathe Bad Air

Drug reactions—multiorgan

DRUG REACTION	CAUSAL AGENTS	NOTES
Antimuscarinic	Atropine, TCAs, H ₁ -blockers, antipsychotics	
Disulfiram-like reaction	1st-generation S ulfonylureas, P rocarbazine, certain C ephalosporins, G riseofulvin, M etronidazole	Sorry Pals, Can't Go Mingle.
Nephrotoxicity/ototoxicity	Aminoglycosides, vancomycin, loop diuretics, cisplatin, amphotericin B	Cisplatin toxicity may respond to amifostine.

Cytochrome P-450 interactions (selected)	Inducers (+)	Substrates	Inhibitors (–)
	Chronic alcohol use St. John's wort Phen ytoin Phen obarbital Nevirapine Rifampin Griseofulvin Carbamazepine	Anti- epileptics Theophylline Warfarin OCPs	Sodium valproate Isoniazid Cimetidine Ketoconazole Fluconazole Acute alcohol abuse Chloramphenicol Erythromycin (macrolides) Sulfonamides Ciprofloxacin Omeprazole Metronidazole
	Chronic alcoholics Steal Phen-Phen and Never Refuse Greasy Carbs	Always Think When Outdoors	SICKFACES.COM

Sulfa drugs

Sulfonamide antibiotics, **Sulfasalazine**, **Probenecid**, **Furosemide**, **Acetazolamide**, **Celecoxib**, **Thiazides**, **Sulfonylureas**. Patients with sulfa allergies may develop fever, urinary tract infection, Stevens-Johnson syndrome, hemolytic anemia, thrombocytopenia, agranulocytosis, acute interstitial nephritis, and urticaria (hives). Symptoms range from mild to life threatening.

Scary Sulfa Pharm FACTS

► PHARMACOLOGY—MISCELLANEOUS

Drug names

ENDING	CATEGORY	EXAMPLE
Antimicrobial		
-azole	Ergosterol synthesis inhibitor	Ketoconazole
-bendazole	Antiparasitic/antihelminthic	Mebendazole
-cillin	Transpeptidase (penicillin-binding protein)	Ampicillin
-cycline	Protein synthesis inhibitor	Tetracycline
-ivir	Neuraminidase inhibitor	Oseltamivir
-navir	Protease inhibitor	Ritonavir
-ovir	DNA polymerase inhibitor	Acyclovir
-thromycin	Macrolide antibiotic	Azithromycin
CNS		
-ane	Inhalational general anesthetic	Halothane
-azine	Typical antipsychotic	Thioridazine
-barbital	Barbiturate	Phenobarbital
-caine	Local anesthetic	Lidocaine
-etine	SSRI	Fluoxetine
-ipramine, -triptyline	TCA	Imipramine, amitriptyline
-triptan	5-HT _{1B/1D} agonist	Sumatriptan
-zepam, -zolam	Benzodiazepine	Diazepam, alprazolam
Autonomic		
-chol	Cholinergic agonist	Bethanechol, carbachol
-curium, -curonium	Nondepolarizing paralytic	Atracurium, vecuronium
-olol	β-blocker	Propranolol
-stigmine	AChE inhibitor	Neostigmine
-terol	β ₂ -agonist	Albuterol
-zosin	α ₁ -antagonist	Prazosin
Cardiovascular		
-afil	PDE-5 inhibitor	Sildenafil
-dipine	Dihydropyridine Ca ²⁺ channel blocker	Amlodipine
-pril	ACE inhibitor	Captopril
-sartan	Angiotensin-II receptor blocker	Losartan
-statin	HMG-CoA reductase inhibitor	Atorvastatin
-xaban	Direct factor Xa inhibitor	Apixaban, edoxaban, rivaroxaban
Other		
-dronate	Bisphosphonate	Alendronate
-glitazone	PPAR-γ activator	Rosiglitazone
-prazole	Proton pump inhibitor	Omeprazole
-prost	Prostaglandin analog	Latanoprost
-tidine	H ₂ -antagonist	Cimetidine
-tinib	Tyrosine kinase inhibitor	Imatinib
-tropin	Pituitary hormone	Somatotropin
-ximab	Chimeric monoclonal Ab	Basiliximab
-zumab	Humanized monoclonal Ab	Daclizumab